

Table 1. Spanish-input, English-output TruthfulQA transfer results using English-calibrated controllers. Repeated rows report mean $\pm$ standard error; compact single-pass rows report means without an error term. TBD marks unrun model-method pairs.

	Method	True (%) $\uparrow$	Informative (%) $\uparrow$	Truthful-concept relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Gemma-2-2B	Original	51.41 $\pm$ 0.90	86.44 $\pm$ 0.58	TBD	1.32 $\pm$ 0.02	1.17 $\pm$ 0.00
	ITI	53.56 $\pm$ 0.47	88.59 $\pm$ 0.45	TBD	1.42 $\pm$ 0.01	1.21 $\pm$ 0.01
	ActAdd	63.57 $\pm$ 0.39	67.15 $\pm$ 0.34	TBD	0.99 $\pm$ 0.01	0.88 $\pm$ 0.01
	Mean-AcT	48.13 $\pm$ 0.52	90.40 $\pm$ 0.46	TBD	1.42 $\pm$ 0.01	<u>1.23 $\pm$ 0.01</u>
	Linear-AcT	47.83 $\pm$ 0.78	90.60 $\pm$ 0.78	TBD	<u>1.44 $\pm$ 0.01</u>	1.23 $\pm$ 0.01
	PID-AcT	48.79 $\pm$ 0.61	<u>90.89 $\pm$ 0.49</u>	TBD	1.43 $\pm$ 0.01	1.22 $\pm$ 0.01
	ODESteer	57.01 $\pm$ 0.20	<u>86.27 $\pm$ 0.28</u>	TBD	1.32 $\pm$ 0.01	1.14 $\pm$ 0.01
	S-PID	55.15 $\pm$ 0.37	86.41 $\pm$ 0.34	TBD	1.29 $\pm$ 0.01	1.12 $\pm$ 0.00
	A-LQR	73.81 $\pm$ 0.71	76.55 $\pm$ 0.65	TBD	1.12 $\pm$ 0.01	1.06 $\pm$ 0.00
	H_∞ (ours)	77.45 $\pm$ 0.52	64.16 $\pm$ 0.85	TBD	1.04 $\pm$ 0.01	1.02 $\pm$ 0.01
Llama-3-8B	Original	42.86 $\pm$ 0.49	94.52 $\pm$ 0.24	TBD	1.59 $\pm$ 0.01	1.39 $\pm$ 0.01
	ITI	55.47 $\pm$ 0.88	94.37 $\pm$ 0.20	TBD	1.61 $\pm$ 0.02	1.39 $\pm$ 0.01
	ActAdd	43.94 $\pm$ 0.53	93.78 $\pm$ 0.29	TBD	1.57 $\pm$ 0.01	1.35 $\pm$ 0.01
	Mean-AcT	46.61 $\pm$ 0.69	94.83 $\pm$ 0.49	TBD	1.61 $\pm$ 0.01	1.41 $\pm$ 0.01
	Linear-AcT	47.15 $\pm$ 0.41	94.96 $\pm$ 0.39	TBD	1.61 $\pm$ 0.01	<u>1.42 $\pm$ 0.02</u>
	PID-AcT	47.12 $\pm$ 0.63	<u>95.08 $\pm$ 0.51</u>	TBD	<u>1.62 $\pm$ 0.01</u>	1.41 $\pm$ 0.01
	ODESteer	<u>95.67 $\pm$ 0.15</u>	11.21 $\pm$ 0.45	TBD	0.00 $\pm$ 0.00	0.02 $\pm$ 0.00
	S-PID	52.09 $\pm$ 0.56	94.37 $\pm$ 0.22	TBD	1.59 $\pm$ 0.01	1.33 $\pm$ 0.01
	A-LQR	55.57 $\pm$ 0.71	95.03 $\pm$ 0.25	TBD	1.56 $\pm$ 0.01	1.27 $\pm$ 0.01
	H_∞ (ours)	85.21 $\pm$ 0.40	64.70 $\pm$ 0.48	TBD	0.97 $\pm$ 0.03	0.97 $\pm$ 0.02
Qwen-2.5-14B	Original	68.13 $\pm$ 0.28	97.31 $\pm$ 0.13	1.82 $\pm$ 0.01	1.82 $\pm$ 0.00	1.51 $\pm$ 0.01
	ITI	67.91 $\pm$ 0.33	97.36 $\pm$ 0.07	1.82 $\pm$ 0.01	1.82 $\pm$ 0.00	1.53 $\pm$ 0.01
	ActAdd	72.09 $\pm$ 0.90	89.62 $\pm$ 1.54	1.66 $\pm$ 0.03	1.63 $\pm$ 0.04	1.37 $\pm$ 0.04
	Mean-AcT	68.27 $\pm$ 0.26	97.48 $\pm$ 0.08	1.82 $\pm$ 0.00	1.82 $\pm$ 0.01	<u>1.55 $\pm$ 0.01</u>
	Linear-AcT	68.25 $\pm$ 0.21	<u>97.55 $\pm$ 0.19</u>	<u>1.83 $\pm$ 0.01</u>	<u>1.83 $\pm$ 0.01</u>	1.55 $\pm$ 0.01
	PID-AcT	68.27 $\pm$ 0.22	97.28 $\pm$ 0.20	1.83 $\pm$ 0.00	1.82 $\pm$ 0.01	1.55 $\pm$ 0.02
	ODESteer	74.88 $\pm$ 0.46	92.83 $\pm$ 0.26	1.73 $\pm$ 0.00	1.67 $\pm$ 0.01	1.44 $\pm$ 0.01
	S-PID	82.82 $\pm$ 0.28	93.56 $\pm$ 0.47	1.74 $\pm$ 0.01	1.64 $\pm$ 0.00	1.43 $\pm$ 0.01
	A-LQR	85.24 $\pm$ 0.50	94.32 $\pm$ 0.27	1.73 $\pm$ 0.01	1.64 $\pm$ 0.01	1.41 $\pm$ 0.01
	H_∞ (ours)	88.74 $\pm$ 0.28	93.05 $\pm$ 0.48	1.65 $\pm$ 0.01	1.53 $\pm$ 0.01	1.29 $\pm$ 0.01
Qwen-2.5-32B	Original	72.86	94.13	1.75	1.68	1.42
	ITI	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD
	S-PID	<u>86.06</u>	89.73	<u>1.71</u>	1.57	1.32
	A-LQR	84.84	<u>92.42</u>	1.69	<u>1.60</u>	<u>1.39</u>
	H_∞ (ours)	84.84	89.24	1.57	1.45	1.30